Auhorized Representative

ThermoFisher SCIENTIFIC

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Submitted to:

Universidade de São Paulo, Escola de Artes, Ciências e Humanidades.

Laboratório de Paleoceanografia e Paleoclima

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Payment Terms	**100% Advanced or Letter of credit**
Payment Mode	**Swift transfer**
Shipment:	**4 months after order**
Delivery Terms:	**FCA, Bremen, Germany**
Quotation Valid Until:	**December 31, 2021**

Thermo Scientific™ MAT 253 Plus Stable Isotopes Mass Spectrometry System

Qty	P/N	Description	Price
1	0723323	MAT 253 Plus Mass Spectrometer, including computer package with ISODAT software suite.	€ 246.095,00
1	BRE0003990	Clumped CO2 collector with four 10E13 ohm resistors for reduced noise level for m/z CO2-47, CO2-48, CO2-49 and including a baseline monitor cup for m/z 47.5.	€ 27.278,00
1	1082863	Dual inlet system including changeover valve for Mat 253 Plus	€ 53.329,00
1	1111122	Differential pump module for Mat 253 Plus	€ 9.742,00
1	0722710	Kiel IV Carbonate preparation device for 253 Plus	€ 122.738,00
1	BRE0023589	Purification trap for clumped isotopes (for Kiel IV Device)	€ 23.460,00
1	1281250	Additional sample magazine incl. 50 vials	€ 4.750,00
1	BRE0000957	Kiel IV Vials (pkg. a 50)	€ 1.506,00
1	1309920	Spare parts and Consumables for KIEL IV	€ 3.808,00
1	0543090	LN2 container (90 L) for carbonate device	€ 11.972,00
1	BRE0013447	Spares & Consumables New Acid Valve for Kiel IV	€ 800,00
1	1322930	Spare parts and consumables (for 253 Plus)	€ 5.582,00
		Total Price, FCA Bremen, Germany	**€ 511.060,00**

Including: Installation.

One Year Warranty factory parts and labor.

Operating Training during installation.

DETAILED DESCRIPTION

0723323 **MAT 253 PLUS** for ultimate precision measurement of the relative isotope abundances of $^{13}C/^{12}C$, $^{15}N/^{14}N$, $^{18}O/^{16}O$, $^{34}S/^{32}S$ (from SO_2 and SF_6), $^{28}Si/^{29}Si$ (from SiF_4) as well as Ar, Kr and Xe, basic instrument consisting of:

1. **Ion Source** – Ultra high sensitivity, self-aligning, gas tight electron impact ion source, optimized for CO_2, CO, N_2, SO_2 as well as for H/D measurements with very low and extremely stable H_3^+ abundance. High ion yield is accomplished by 10 kV accelerating voltage and Einzel lens focusing. Ion source parameters are set manually or automatically through the data system.

2. **Ion Source Housing** - Stainless steel monoblock design without any welding seams to guarantee clean and smooth surfaces for memory-free small sample measurements.

3. **Vacuum System** - Computer-controlled, fail-safe high vacuum system with a TMH 262 turbomolecular pump that incorporates a Hollweck stage for improved light gas pumping and a 5 l/s rotary forevacuum pump. The vacuum system is all-metal sealed and provides ultrahigh vacuum conditions. No cooling water is required. Automatic system protection in case of power and vacuum failure. The instrument control panel provides display of high vacuum status in the analyzer and optional dual inlet system.

4. **VISC** - **V**ariable **I**on **S**ource **C**onductance easily adjustable from outside the vacuum. VISC allows optimizing changeover times and avoids memory effects for e.g. high precision SO_2 measurements. It also optimizes the gas flow characteristics for continuous flow isotope ratio monitoring.

5. **Ion Optics** – 90° stigmatic focusing magnetic sector type with near to 100 % transmission. 46 cm equivalent deflection radius for N_2, CO_2, SO_2 and 18 cm radius for H/D. 10 kV computer-controlled acceleration potential for full mass range and 1.0 Tesla electromagnet for mass separation. Magnet field control by digital to analog converter with 16 bit resolution. Acceleration voltage and magnetic field are controlled by the data system.

   High mass range capability at full 10 kV accelerating voltage up to 150 u for analysis of e. g. SF_6 or SiF_4 samples (optional cup configuration required). Extended mass range at decreased accelerating voltage by is accessible ISODAT control.

6. **Electronics** - Modular design concept using an internal bus structure. The electronic modules are under computer control via a specially designed light pipe interface for total elimination of galvanic coupling effects and grounding loops.

   - All ion source parameters can be set by the computer. The ion source parameters can be automatically optimized, stored, loaded and appended to specific gas configurations.
   - All analyzer parameters are controlled by the data system.
   - Read back of system status parameters such as accelerating voltage, emission current, magnet current and vacuum pressures.

7. **Universal CNOS collector system** – comprising five Faraday cups for simultaneous collection of ion currents, e.g. CO_2 (44, 45, 46), O_2 (32, 33, 34), N_2 (32, 33, 34), CO (28, 29, 30), NO (30, 31, 32) or SO_2 (64, 66) can be detected in static mode.

   - Highly stable low-noise amplifiers (FET's) are used for the ion current measurements.
   - Amplifiers with linear dynamic range of 50 V.
   - Amplification factors can be optionally switched under computer control (see options PN 0723520, 0723530, 0723540).
   - Amplifiers are mounted in a separate housing kept under vacuum to minimize effects from changing environmental conditions. All amplifiers are designed for fast data acquisition. High ohmic resistors are adapted to the isotope abundances.
   - Up to eight integrating channels are available for simultaneous ion current measurement upon request (see option # 0722360 additional Faraday channel for the expansion of the Universal CNOS collector). The required Faraday cup arrangement must be specified with order.

8. **Isodat Software Suite**

   - The Isodat software suite is a seamlessly integrated part of the DELTA V, including its peripherals, controlling, and automating the complete analysis from sample preparation to data reporting.
   - Automated analysis is enabled by a straightforward method and sequence set-up.
   - Automated user-tailored reporting and flexible data export including all network capabilities.
   - Complex experiments are supported by full control of operation parameters.
   - Integrated scrip language “ISL” enables the experienced scientist to take complete control of all aspects of the IRMS, including standard peripherals and home-build preparation and inlet devices.
   - Unique package of diagnostic functions checking all important aspects of the mass spectrometer.
   - Multiple user license for data evaluation.

9. **Data System**

Personal Computer - minimum configuration:

OptiPlex 7050 Tower or better

- 33454279 - OptiPlex 7050 Minitower XCTO
- 1 x 329-BDHN OptiPlex 7050 MT with 240W 80PLUS Bronze)
- 1 x 338-BHUF Intel Core i5-6500 QC/6MB/4T/3.2GHz/65W); supports Windows 7/8.1/10/Linux (or better)
- 1 x 325-BBRL DVD+/-RW
- 1 x 370-ADJS 16GB (2x8GB) 2.400MHz DDR4
- 1 x 400-AHZI 2,5 Zoll 1TB 7.200 1/min
- 1 x 429-AAJV Tray Loading Dual Layer DVD Burner
- 1 x 389-BHJT Intel Core i5 vPro Processor Label
- 1 x 490-BBFG Intel Integrated Graphics,OptiPlex
- 1 x 382-BBDO VGA video port, Mini Tower
- 1 x 382-BBDJ PCIe card with serial and parallel port,Full Height
- 1 x 570-AAIQ Dell Optical Mouse MS116 Black
- 1 x 580-ADEB Dell Multimedia Keyboard KB216 Black (US International - QWERTY)
- 1 x 709-12335 Base Warranty
- 1 x 709-12336 3Yr Basic Warranty - Next Business Day - Minimum Warranty
- 1 x 710-45869 3Yr ProSupport and Next Business Day On-Site Service
- High resolution 22-inch TFT color monitor

10. **Installation Specifications**

The following analytical standard specifications will be performed during MAT 253 PLUS installation. The successful demonstration is the basis for system acceptance.

- **Absolute Sensitivity**

  600 molecules CO_2 per m/z 44 ion under normal Dual Inlet operating conditions

- **Sensitivity under Continuous Flow Conditions**

  900 molecules CO_2 per m/z 44 ion with He load will guarantee an isotope ratio linearity better than 0.02 ‰/nA

- **System Stability**

  < 10 ppm on mass scale

- **H_3-Factor**

  < 10 ppm/nA

- **Reference Gas Precision ***

  ^{13}C (CO_2) 0.06 ‰ for 5 nA signal at 1.5 V (SD 1 σ for 10 consecutive pulses)

  ^{18}O (CO_2) 0.08 ‰ for 5 nA signal at 1.5 V (SD 1 σ for 10 consecutive pulses)

  measured under Continuous Flow conditions

- **Reference Gas Linearity ***

  ^{13}C (CO_2) 0.15 ‰ SD 1 σ for 2 – 20 nA

  ^{13}C (CO_2) 0.02 ‰/nA for 2 – 60 nA

  * Only in case a continuous flow peripheral is installed.

BRE0003990 **Clumped collector system** – Clumped CO2 collector with four 10E13 ohm resistors with reduced noise level for m/z CO2-47, CO2-48, CO2-49 incl. baseline monitor cup Highly stable low-noise amplifiers (FET's) are used for the ion current measurements.

1082863 **Dual Inlet System** - Viscous gas flow dual inlet system with variable volume bellows for sample and reference gas (3 - 40 mL). Membrane pressure transducers on both variable volumes for reference gas and sample gas monitoring. Automated pressure adjustment by computer control.

The two-port changeover valve is mounted directly onto the ion source to ensure short gas connection to the ionization chamber of the electron impact ion source.

Gas transfer to the changeover valve is via two capillaries (with adjustable crimp leaks) ensuring viscous flow.All valves of the inlet system are of monoblock design for low dead volume and small surface areas to guarantee high conductance and to avoid memory and cross mixing.

Valve seats have gold gaskets and gold stem tips. The dual inlet system also includes the capillaries for reference and sample gas as well as the changeover valve.Vacuum is maintained by a turbomolecular pump and 1.5 L/s rotary fore vacuum pump for pumping the waste line and for high vacuum pumping of the inlet system. The inlet system is heatable to 80° C (e.g. for SO2 measurements).

1111122 Differential pump module for Mat 253 Plus with turbomolecular pump.

0722710 **Kiel IV Carbonate Preparation Device** - with individual acid dose per sample for DELTA V series.

Carbonate sample preparation device with acid dosing valve and 46 separate sample vials for individual acid reaction of each sample. The system is fully automatic, and microprocessor controlled.

The Kiel IV carbonate device requires a 90 L container for liquid N_2 (PN 0543090) and Dual Inlet system installed. A differential pumping system is required for DELTA+/XL series (PN 1082810) and MAT 252 (PN 1111120) instruments.

Installation Standard Specifications

The following analytical standard specifications will be performed during installation. The successful demonstration is the basis for system acceptance:

Standard External Precision

^{13}C (CO_2) 0.04 ‰ at sample size >20 µg

^{18}O (CO_2) 0.08 ‰ at sample size >20 µg

BRE0023589 Purification Device for clumped isotopes

1281250 Additional Sample Magazine (46 samples) for Kiel IV Carbonate Preparation Device.

BRE0000957 Kiel IV Vials (pkg. a 50).

1309920 Spare parts and Consumables for Kiel IV Carbonate Preparation Device.

0543090 Liquid N_2-Container for Kiel IV carbonate device - Automatic liquid nitrogen supply (90 L) for cryogenic trapping system.

BRE0013447 Spare & Consumables for KIEL IV unit for acid valve parts

1322930 Spare Parts and Consumables, recommended set of spares and consumables for MAT 253 basic unit for approximately one (1) year of operation.

Please note:

Certified reference materials for on-site measurements by Thermo Fisher Scientific service personnel must be provided by the customer. Any deviations from standard specifications are only valid upon written approval in each case. If extra performance specifications are agreed on, they will be demonstrated and documented only once, either in the factory in the test field, or on-site during installation. If the special specifications need to be shown twice (in the factory and on-site), they need to be ordered twice. Material is included unless the special specifications are to be demonstrated on-site. In this case, the samples must be delivered by the user.

Please Note: Site preparation/supplies such as carrier gas, electricity, air conditioning, are the purchaser's responsibilities. These preparation requirements must be met for warranty to remain valid. Specifications are available upon request. Please check our Pre-installation Guide.

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1. GENERAL. Thermo Electron North America LLC (" **Seller**") hereby offers for the products the express condition that Buyer agrees to accept and be bound by the terms and conditions set forth herein. Any new or different provisions contained in any document issued by Buyer in response to this offer are expressly rejected.

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3. TERMS OF PAYMENT. Seller may invoice Buyer when requested and according to payment terms as per Advanced Payment or Irrevocable and confirmed Letter of Credit.

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ANY INSTALLATION, MAINTENANCE, REPAIR, SERVICE, RELOCATION, OR ALTER ATION TO OR OF, OR OTHER TAMPERING WITH, THE PRODUCTS PERFORMED BY ANY PERSON OR ENTITY OTHER THAN SELLER WITHOUT SELLER'S PRIOR WRITTEN APPROVAL, OR ANY USE OF REPLAC EMENT PARTS NOT S UPPLIED BY SELLER, SHALL IMMEDIATELY VOID AND CANCEL ALL WARRANTIES WITH RESPECT TO THE AFFECTED PRODUCTS AND/OR SERVICES. THE OBLIGATIONS CREATED BY THIS WARRANTY

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for which they were not designed; (b) improper storage and handling of the Products; (c) use of the Products in combination with equipment or software not supplied by Seller (d) shipping damage incurred en route to Buyer's site or because of moving equipment, in which case Seller will promptly provide a cost estimate for Technical Support to the consignee for filing claims to carriers for shipping damage; (e) flood, lightning, earthquake, tornado, hurricane or fire, bombing, armed conflict, malicious mischief, sabotage or other natural or man-made disasters; (f) normal wear and tear, physical abuse, misuse, sprinkler damage, electrical surge or abnormal power variation; (g) repairs, maintenance, or modifications made by anyone other than Seller trained personnel or without Seller's supervision and/ or approval; (h) relocation and reinstallation of equipment; although upon request Seller will supervise the removing, crating, relocation and reinstallation of the Products at Seller's current Services rates; (i) maintenance or replacement of media (i.e., floppy disks, plotter supplies, etc.) whatever the reason f or loss, failure or damage; (j) b eta-site support; (k) operator training; or (l) repairing Product malfunctions if the fault is not with the equipment. If Seller determines that Products for which Buyer

standard performance specifications. Manuals: Delivery of One set of complete operator and maintenance manuals.

Note: Site preparation/supplies such as carrier gas, electricity and air conditioning are the purchaser's responsibility. These preparation requirements must be met for warranty to remain valid. (Specifications are available upon request.)

Banking data:

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Bank of America

Address: New York, NY

Account Number: 4426395159

ABA: 026009593 SWIFT: BOFAUS3N